

Titanic Dataset Exploratory Data Analysis

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_style("whitegrid")
```

Load Dataset

```
import pandas as pd

df = pd.read_csv("Titanic-Dataset.csv")
df.head()
```

	PassengerId	Survived	Pclass	\
0	1	0	3	
1	2	1	1	
2	3	1	3	
3	4	1	1	
4	5	0	3	

	Name	Sex	Age
SibSp \			
0	Braund, Mr. Owen Harris	male	22.0
1			
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0
1			
2	Heikkinen, Miss. Laina	female	26.0
0			
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0
1			
4	Allen, Mr. William Henry	male	35.0
0			

	Parch	Ticket	Fare	Cabin	Embarked
0	0	A/5 21171	7.2500	NaN	S
1	0	PC 17599	71.2833	C85	C
2	0	STON/O2. 3101282	7.9250	NaN	S
3	0	113803	53.1000	C123	S
4	0	373450	8.0500	NaN	S

Dataset Information

```
df.info()
```

```

<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  -
0   PassengerId      891 non-null    int64
1   Survived         891 non-null    int64
2   Pclass           891 non-null    int64
3   Name             891 non-null    str
4   Sex              891 non-null    str
5   Age              714 non-null    float64
6   SibSp            891 non-null    int64
7   Parch            891 non-null    int64
8   Ticket           891 non-null    str
9   Fare             891 non-null    float64
10  Cabin            204 non-null    str
11  Embarked         889 non-null    str
dtypes: float64(2), int64(5), str(5)
memory usage: 83.7 KB

```

Statistical Summary

```
df.describe()
```

	PassengerId	Survived	Pclass	Age	SibSp	\
count	891.000000	891.000000	891.000000	714.000000	891.000000	
mean	446.000000	0.383838	2.308642	29.699118	0.523008	
std	257.353842	0.486592	0.836071	14.526497	1.102743	
min	1.000000	0.000000	1.000000	0.420000	0.000000	
25%	223.500000	0.000000	2.000000	20.125000	0.000000	
50%	446.000000	0.000000	3.000000	28.000000	0.000000	
75%	668.500000	1.000000	3.000000	38.000000	1.000000	
max	891.000000	1.000000	3.000000	80.000000	8.000000	

	Parch	Fare
count	891.000000	891.000000
mean	0.381594	32.204208
std	0.806057	49.693429
min	0.000000	0.000000
25%	0.000000	7.910400
50%	0.000000	14.454200
75%	0.000000	31.000000
max	6.000000	512.329200

Missing Values

```
df.isnull().sum()
```

```
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
Age          177
SibSp         0
Parch         0
Ticket        0
Fare          0
Cabin        687
Embarked       2
dtype: int64
```

Unique Values

```
df.nunique()

PassengerId    891
Survived        2
Pclass         3
Name           891
Sex            2
Age           88
SibSp          7
Parch          7
Ticket        681
Fare          248
Cabin         147
Embarked        3
dtype: int64
```

Value Counts

```
df["Sex"].value_counts()

df["Pclass"].value_counts()

df["Survived"].value_counts()

Survived
0      549
1      342
Name: count, dtype: int64
```

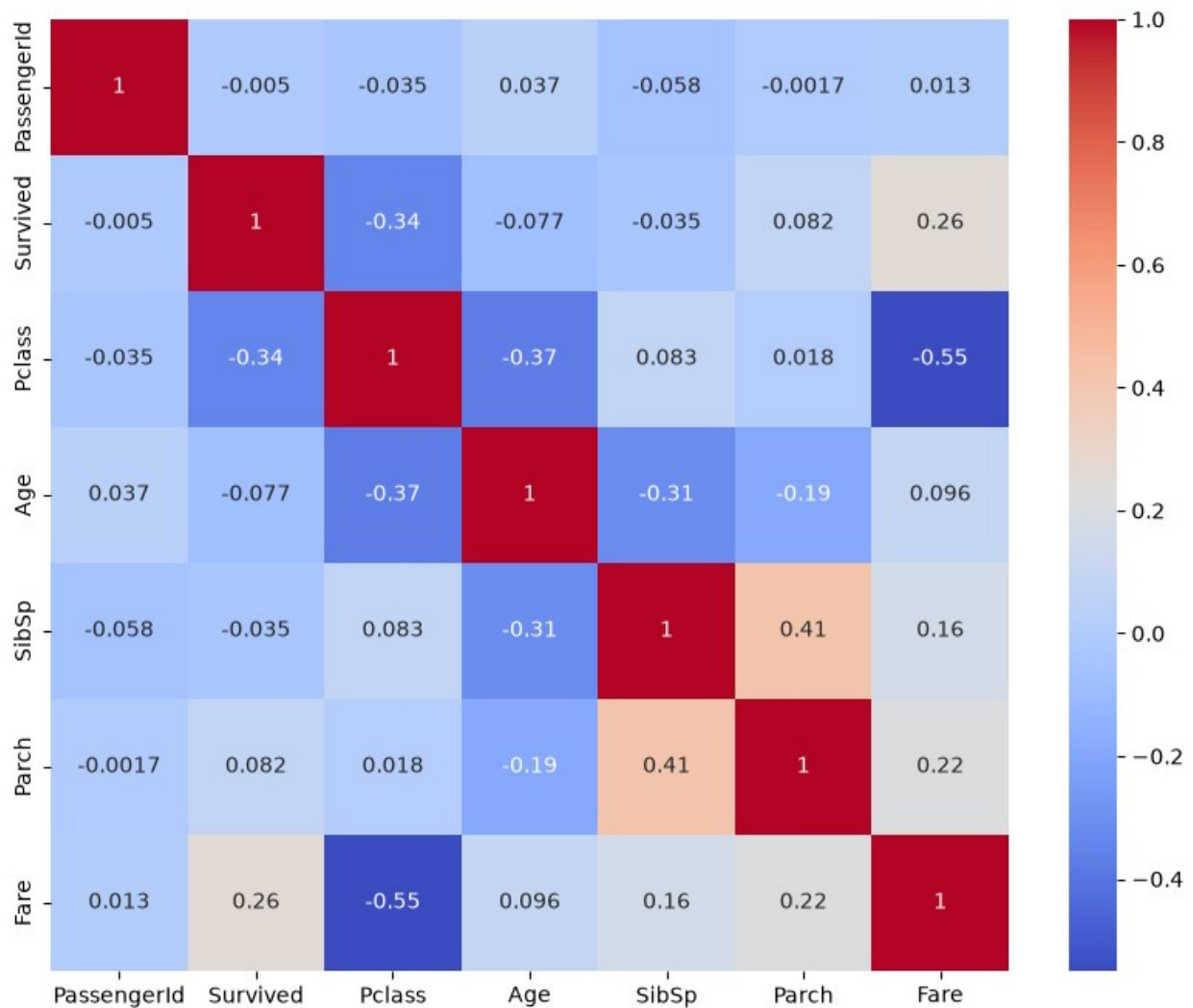
Correlation Heatmap

```
import matplotlib.pyplot as plt
import seaborn as sns
```

```
plt.figure(figsize=(10,8))

sns.heatmap(
    df.corr(numeric_only=True),
    annot=True,
    cmap="coolwarm"
)

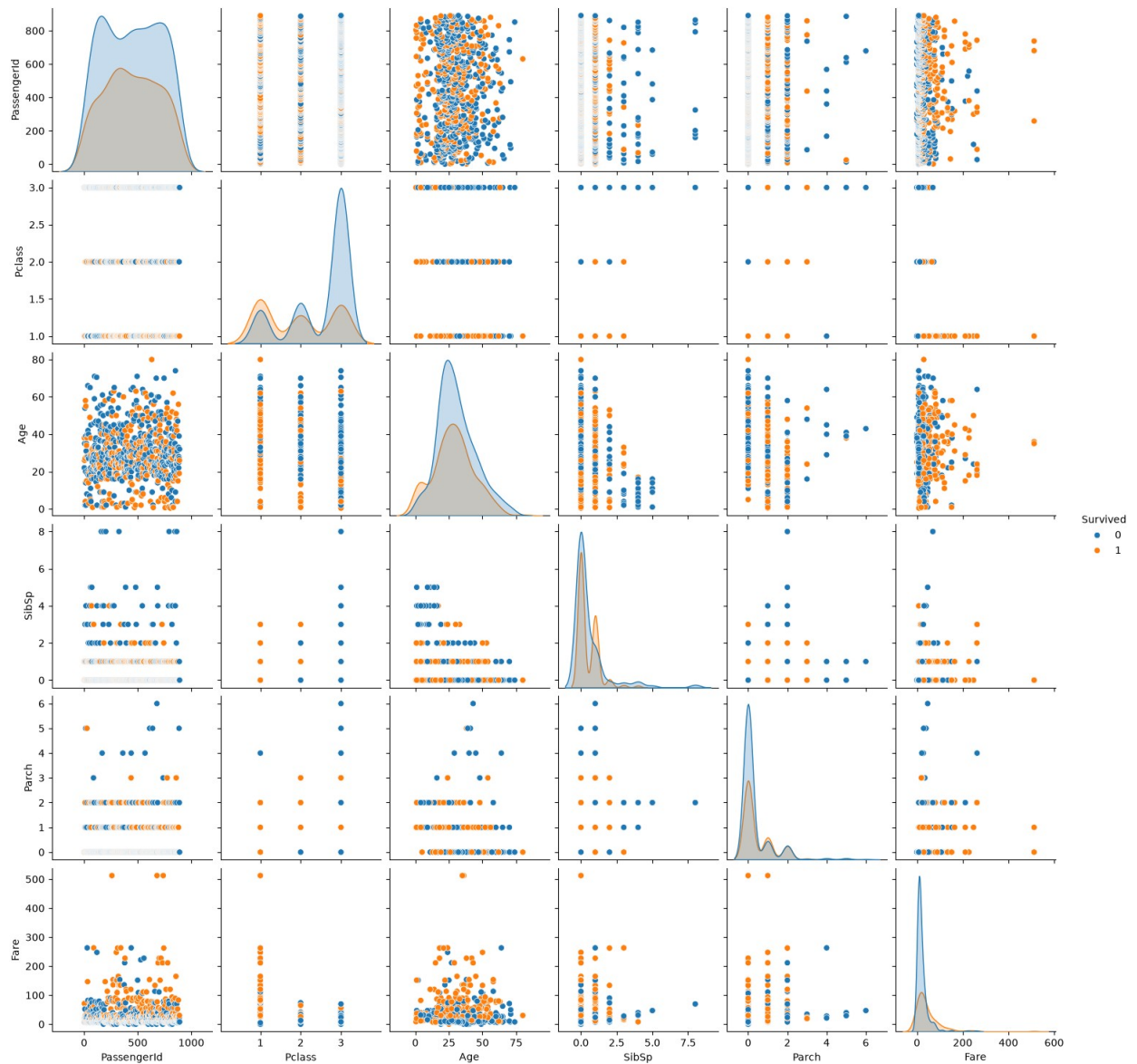
plt.show()
```



Pairplot

```
sns.pairplot(df,
    hue="Survived")

plt.show()
```

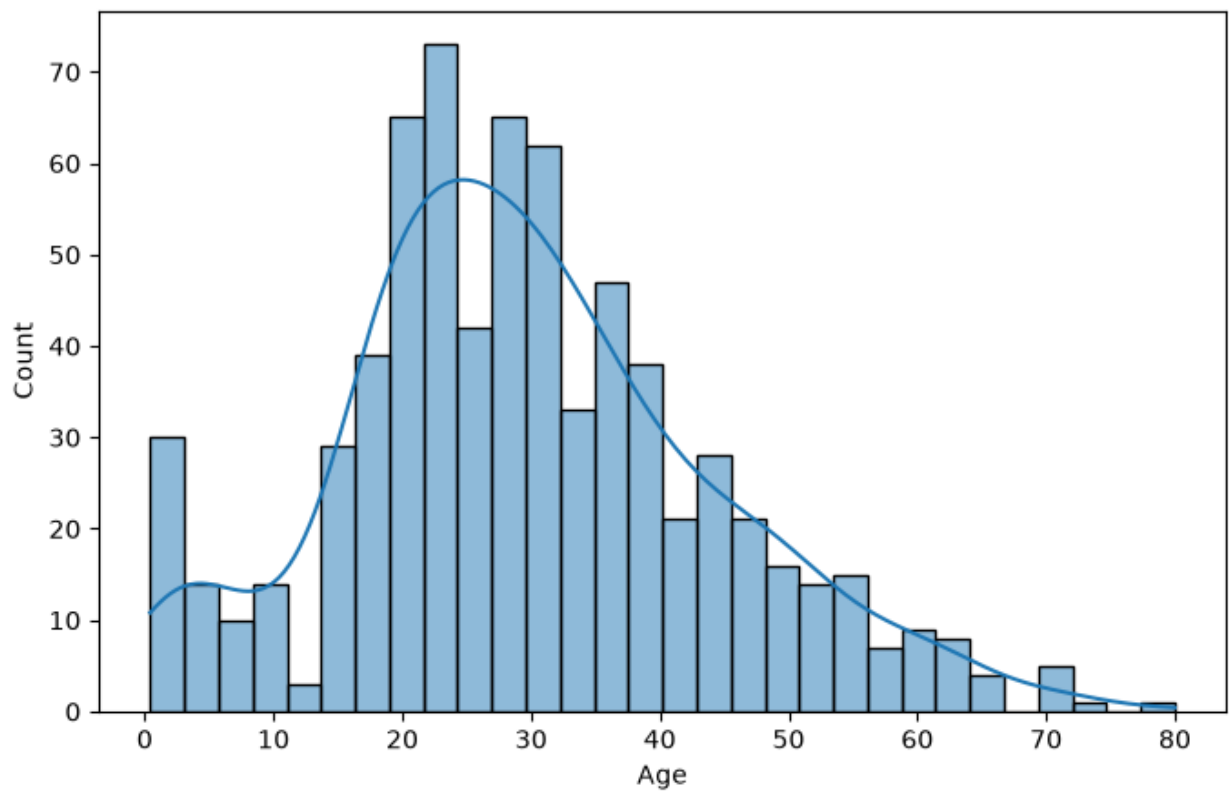


Age Distribution

```
plt.figure(figsize=(8,5))

sns.histplot(df["Age"],
             bins=30,
             kde=True)

plt.show()
```

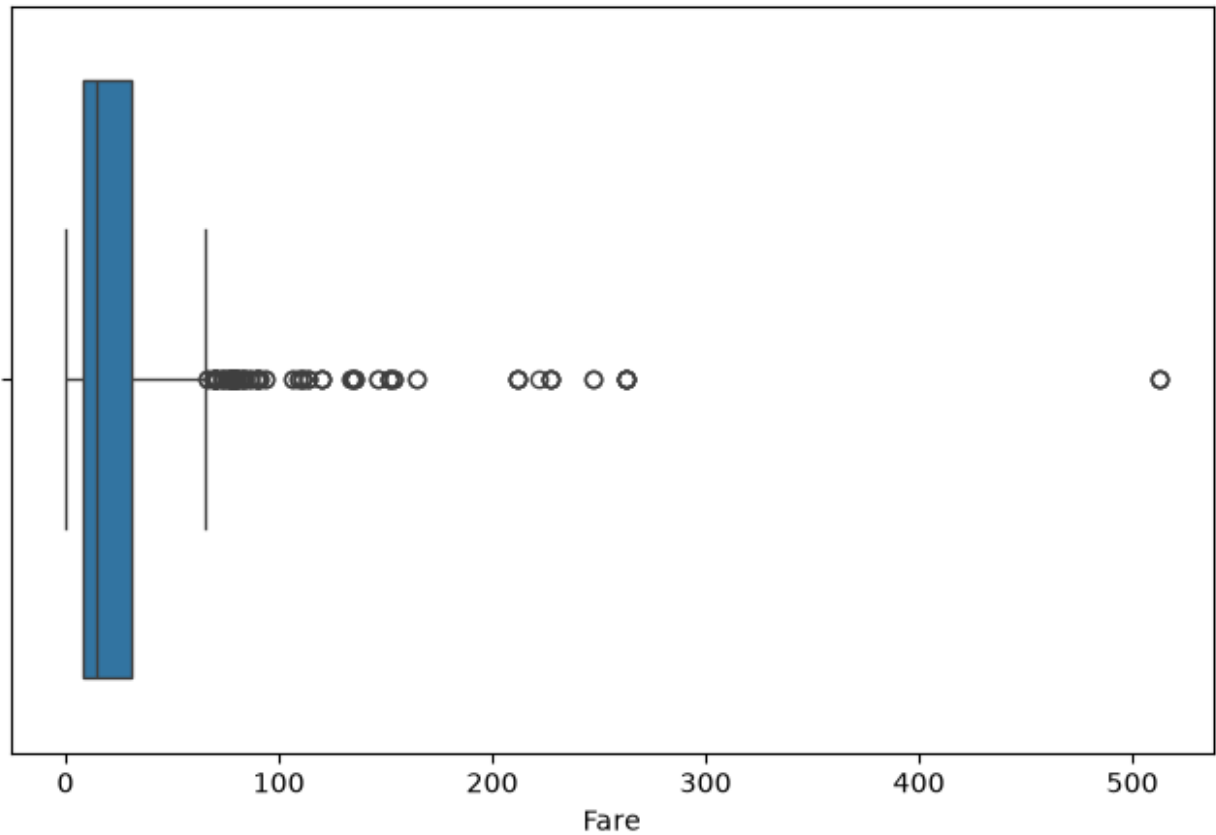


```
## Fare Distribution
```

```
plt.figure(figsize=(8,5))
```

```
sns.boxplot(x=df["Fare"])
```

```
plt.show()
```

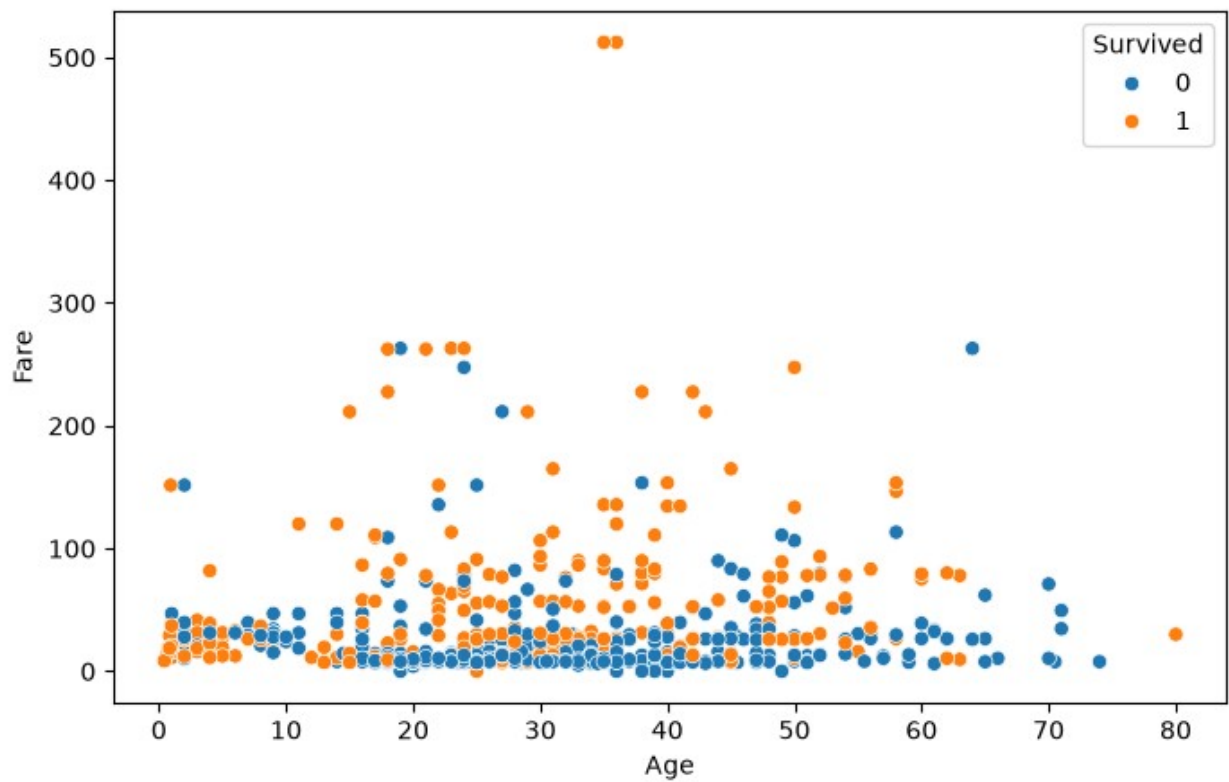


Age vs Fare

```
plt.figure(figsize=(8,5))

sns.scatterplot(
    x="Age",
    y="Fare",
    hue="Survived",
    data=df
)

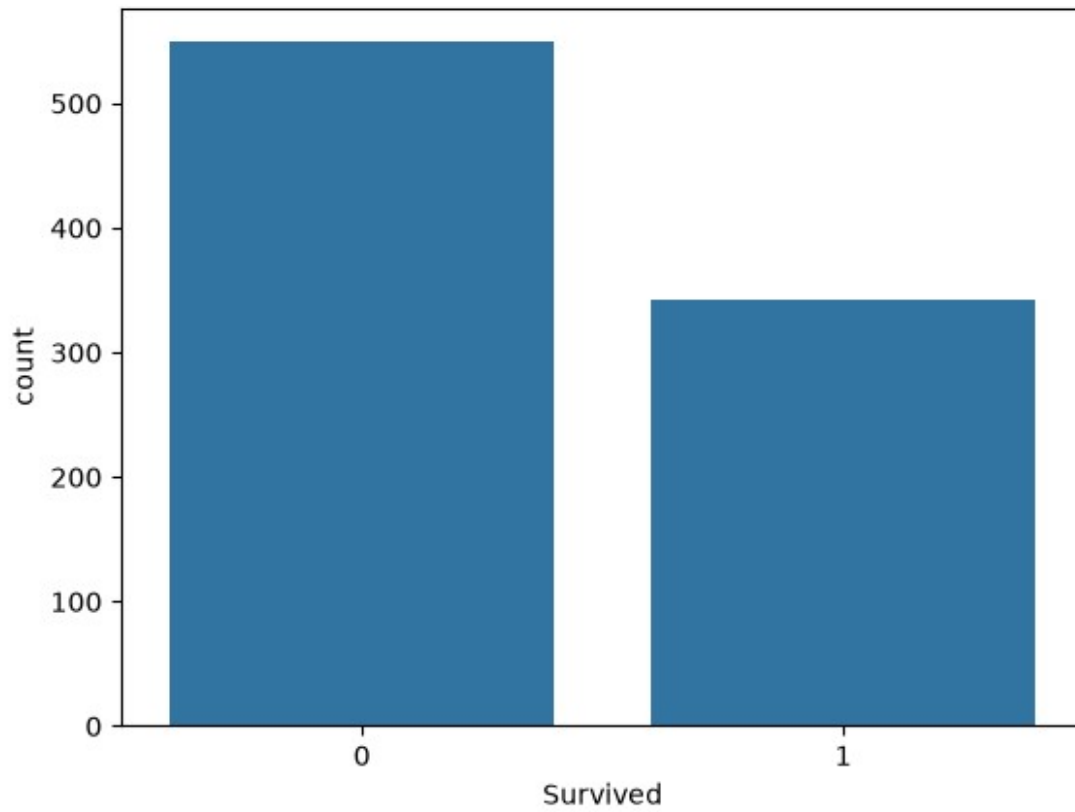
plt.show()
```



```
## Survival Count
```

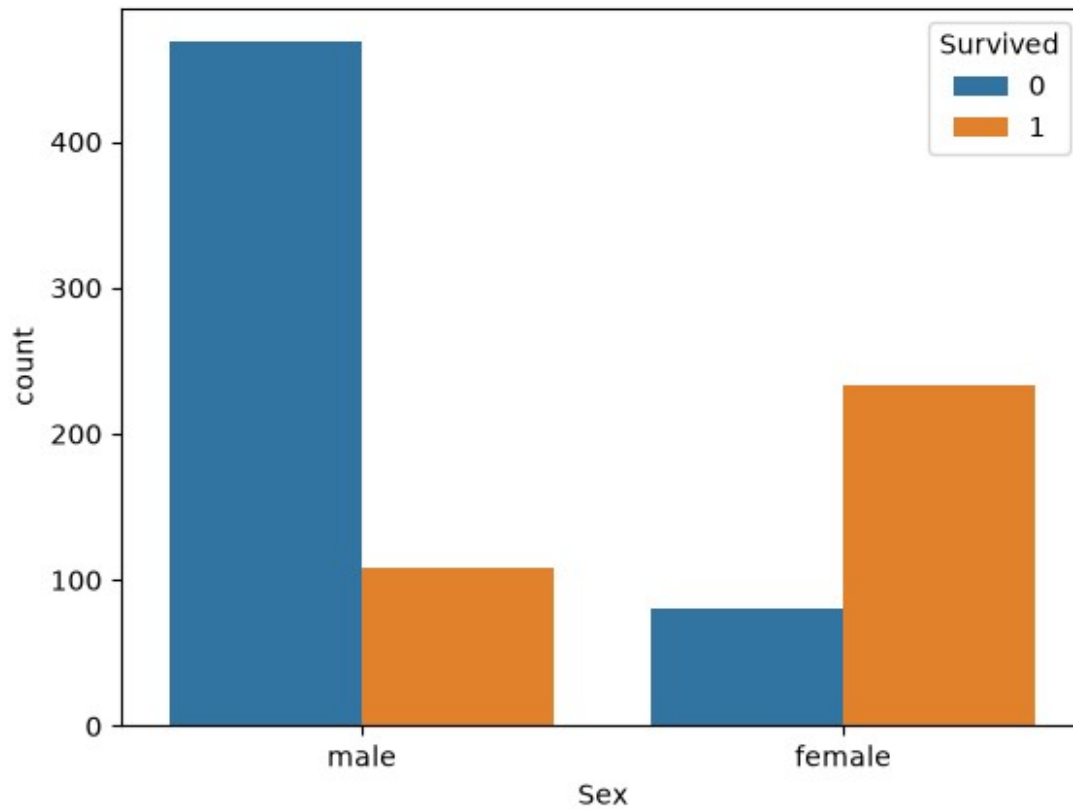
```
sns.countplot(  
    x="Survived",  
    data=df  
)
```

```
plt.show()
```

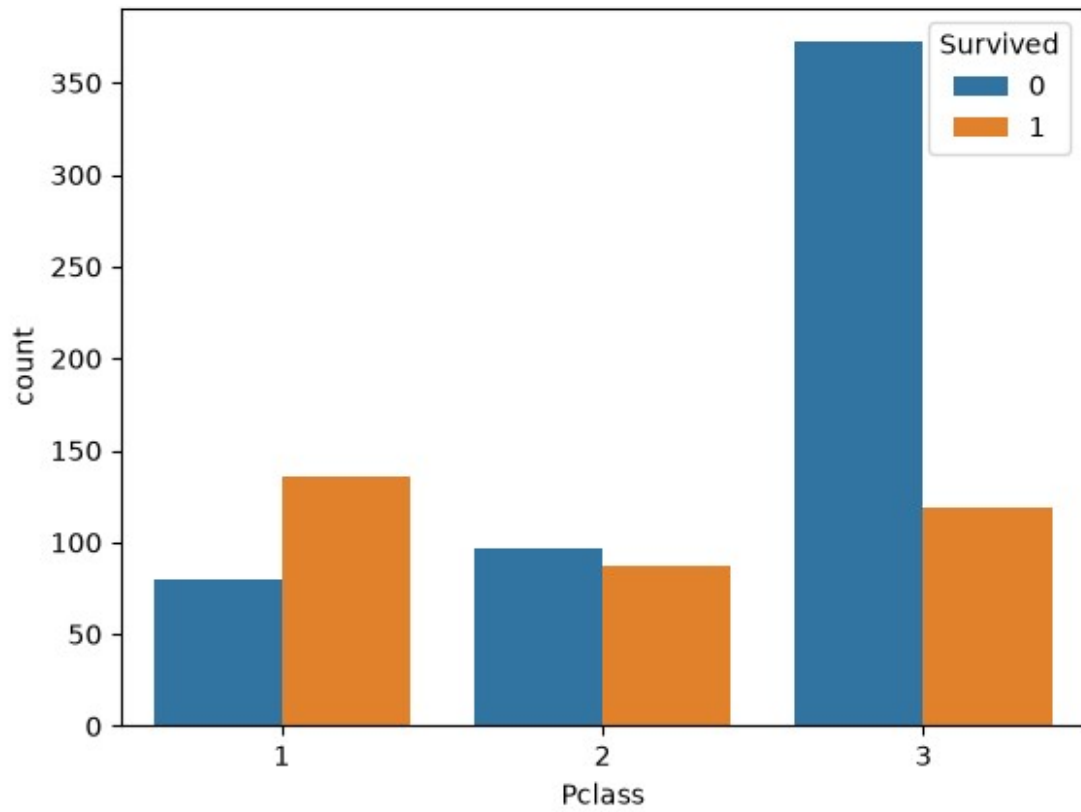
Survival by Gender

```
sns.countplot(  
    x="Sex",  
    hue="Survived",  
    data=df  
)  
plt.show()
```



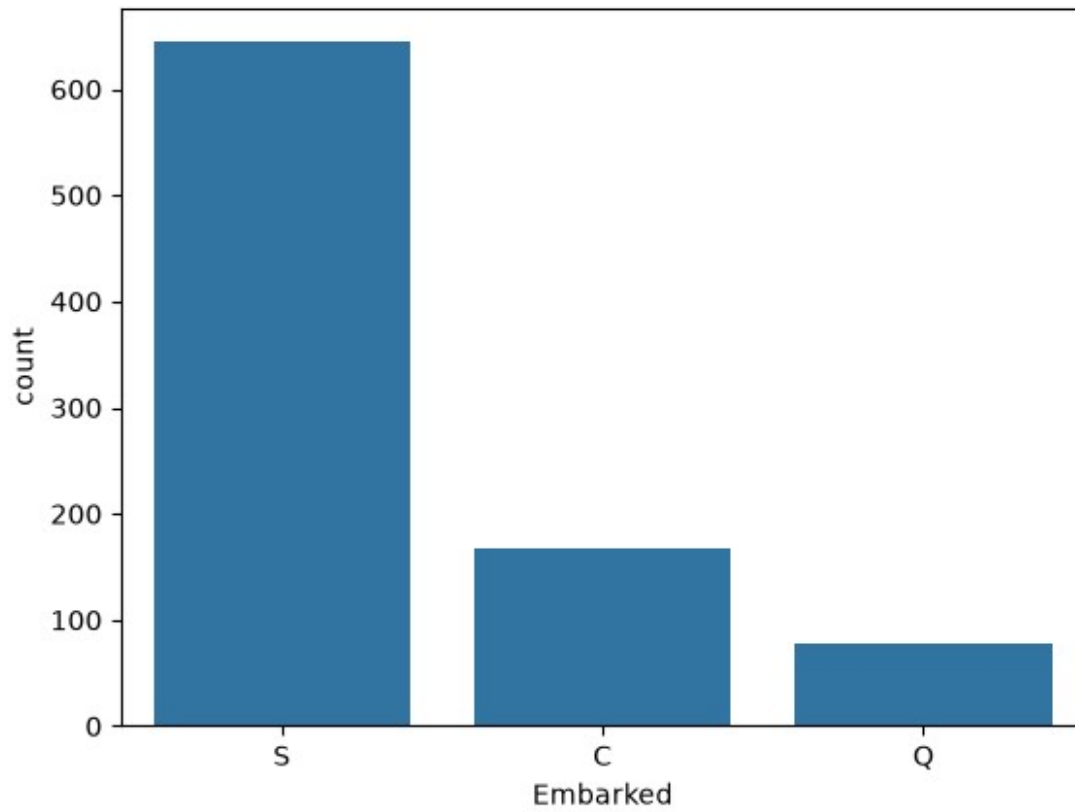
Passenger Class Distribution

```
sns.countplot(  
    x="Pclass",  
    hue="Survived",  
    data=df  
)  
plt.show()
```



Embarked Port

```
sns.countplot(  
    x="Embarked",  
    data=df  
)  
plt.show()
```

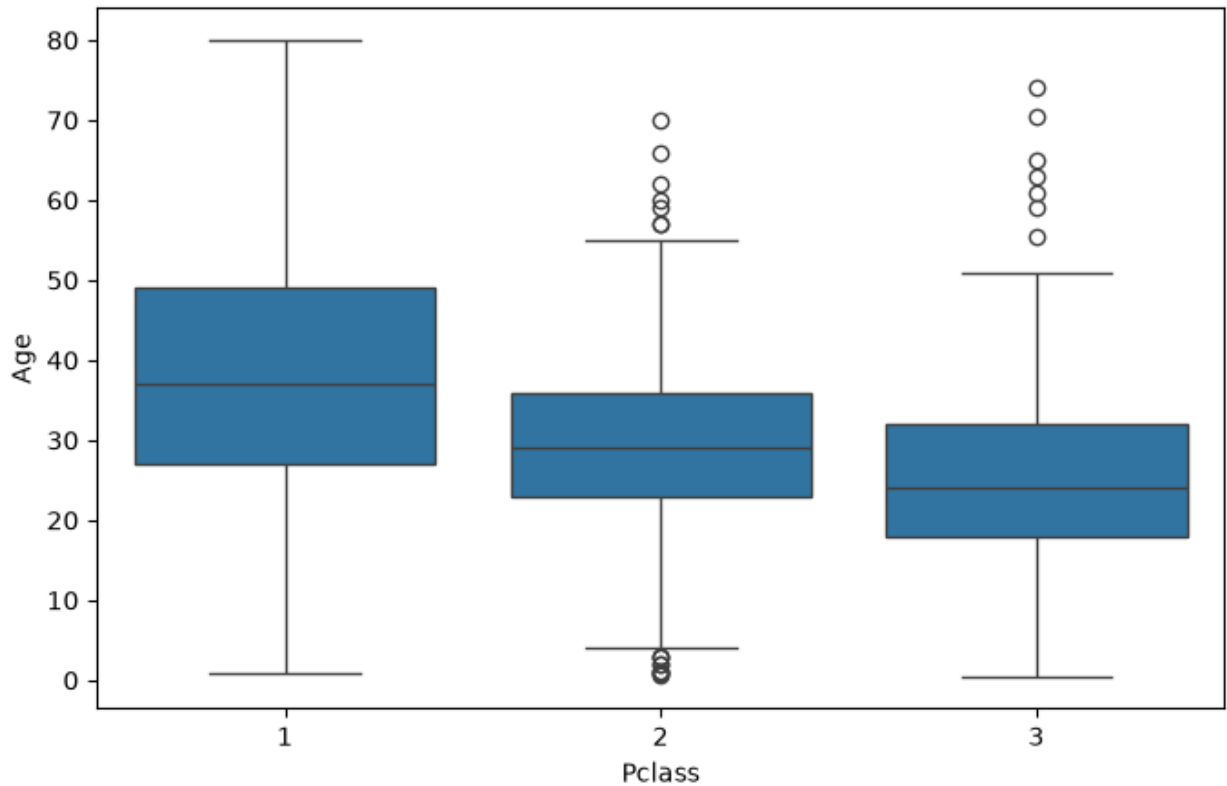


Age by Passenger Class

```
plt.figure(figsize=(8,5))
```

```
sns.boxplot(  
    x="Pclass",  
    y="Age",  
    data=df  
)
```

```
plt.show()
```



Summary of Findings

1. Females had a much higher survival rate than males.
2. First-class passengers survived more frequently.
3. Fare has a positive relationship with survival.
4. Most passengers were between 20–40 years old.
5. Fare contains several outliers.
6. Cabin has many missing values.
7. Age has missing values.
8. Most passengers embarked from Southampton.